Orthogonality and Gram–Schmidt

Problem sheet for the Day 9 class · CSIR NET / GATE previous year questions

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. GATE 2011 MCQ The application of Gram-Schmidt process of orthonormalization to $u_1 = (1,1,0),\ u_2 = (1,0,0),\ u_3 = (1,1,1)$ yields

(A) $\frac{1}{\sqrt{2}}(1,1,0),\ (1,0,0),\ (0,0,1)$

(B) $\frac{1}{\sqrt{2}}(1,1,0),\ \frac{1}{\sqrt{2}}(1,-1,0),\ \frac{1}{\sqrt{2}}(1,1,1)$

(C) $(0,1,0),\ (1,0,0),\ (0,0,1)$

(D) $\frac{1}{\sqrt{2}}(1,1,0),\ \frac{1}{\sqrt{2}}(1,-1,0),\ (0,0,1)$

2. GATE 2011 MCQ Let $T : \mathbb{C}^3 \to \mathbb{C}^3$ be defined by $T(z_1,z_2,z_3) = (z_1 - iz_2,\ iz_1 + z_2,\ z_1 + z_2 + iz_3)$. Then the adjoint T^* of T is given by $T^*(z_1,z_2,z_3) =$

(A) $(z_1 + iz_2,\ -iz_1 + z_2,\ z_1 + z_2 - iz_3)$

(B) $(z_1 - iz_2 + z_3,\ -iz_1 + z_2 + z_3,\ iz_3)$

(C) $(z_1 - iz_2 + z_3,\ iz_1 + z_2 + z_3,\ -iz_3)$

(D) $(iz_1 + z_2,\ z_1 - iz_2,\ z_1 - z_2 - iz_3)$

3. GATE 2013 MCQ Let $\mathcal{H}$ be a Hilbert space and let $\{e_n : n \geq 1\}$ be an orthonormal basis of $\mathcal{H}$. Suppose $T : \mathcal{H} \to \mathcal{H}$ is a bounded linear operator. Which of the following CANNOT be true?

(A) $T(e_n) = e_1$ for all $n \geq 1$

(B) $T(e_n) = e_{n+1}$ for all $n \geq 1$

(C) $T(e_n) = \sqrt{\frac{n+1}{n}}\, e_n$ for all $n \geq 1$

(D) $T(e_n) = e_{n-1}$ for all $n \geq 2$ and $T(e_1) = 0$

4. GATE 2012 MCQ Let H be a complex Hilbert space, $T : H \to H$ be a bounded linear operator and let T^* denote the adjoint of T. Which of the following statements are always TRUE? P: $\langle Tx, y\rangle = \langle x, T^*y\rangle\ \forall x, y$. Q: $\langle x, Ty\rangle = \langle T^*x, y\rangle\ \forall x, y$. R: $\langle x, Ty\rangle = \langle x, T^*y\rangle\ \forall x, y$. S: $\langle Tx, Ty\rangle = \langle T^*x, T^*y\rangle\ \forall x, y$.

(A) P and Q

(B) P and R

(C) Q and S

(D) P and S

5. GATE 2012 MCQ For the matrix $M = \begin{pmatrix} 2 & 3+2i & -4 \\ 3-2i & 5 & 6i \\ -4 & -6i & 3 \end{pmatrix}$, which of the following statements are correct? P: M is skew-Hermitian and iM is Hermitian. Q: M is Hermitian and iM is skew-Hermitian. R: eigenvalues of M are real. S: eigenvalues of iM are real.

(A) P and R only
(B) Q and R only
(C) P and S only
(D) Q and S only

6. CSIR NET June 2014 MSQ Let u, v, w be vectors in an inner-product space V, satisfying $\|u\| = \|v\| = \|w\| = 2$ and $\langle u, v\rangle = 0,\ \langle u, w\rangle = 1,\ \langle v, w\rangle = -1$. Then which of the following are true?

(A) $\|w + v - u\| = 2\sqrt{2}$
(B) $\{\frac{1}{2}u, \frac{1}{2}v\}$ forms an orthonormal basis of a two dimensional subspace of V
(C) w and $4u - w$ are orthogonal to each other
(D) u, v, w are necessarily linearly independent

7. CSIR NET Dec 2013 MSQ Let y be a nonzero vector in an inner product space V. Then which of the following are subspaces of V?

(A) $\{x \in V \mid \langle x, y\rangle = 0\}$
(B) $\{x \in V \mid \langle x, y\rangle = 1\}$
(C) $\{x \in V \mid \langle x, z\rangle = 0 \text{ for all } z \text{ such that } \langle z, y\rangle = 0\}$
(D) $\{x \in V \mid \langle x, z\rangle = 1 \text{ for all } z \text{ such that } \langle z, y\rangle = 1\}$

8. CSIR NET June 2015 MSQ Let u be a real $n \times 1$ vector with $u'u = 1$ (u' the transpose). Define $A = I - 2uu'$, $I = I_n$. Which of the following statements are true?

(A) A is singular
(B) $A^2 = A$
(C) $\text{Trace}(A) = n - 2$
(D) $A^2 = I$

9. CSIR NET June 2018 MSQ Let V be an inner product space and S be a subset of V. Let $\bar{S}$ denote the closure of S in the norm topology. Which of the following statements are true?

(A) $S = (S^{\perp})^{\perp}$
(B) $\bar{S} = (S^{\perp})^{\perp}$
(C) $\overline{\text{span}(S)} = (S^{\perp})^{\perp}$
(D) $S^{\perp} = ((S^{\perp})^{\perp})^{\perp}$

10. CSIR NET June 2017 MSQ Let m, n, r be natural numbers. Let A be an $m \times n$ matrix with real entries such that $(AA^t)^r = I$, where I is the $m \times m$ identity. We can conclude that

(A) $m = n$
(B) AA^t is invertible
(C) A^tA is invertible
(D) if $m = n$, then A is invertible